

KEY TECHNICAL SKILLS

JavaScript/Typescript
Python
HTML/CSS
NodeJS
React
React Native
Django
Docker
Terraform
MongoDB/DynamoDB
PostgreSQL
AWS stack
GitHub Actions
Bitbucket/Jira
DevOps
Jest/Vitest Testing
Playwright E2E Testing
AI/LLM Engineering

CONTACT

0422 937 023
chapmanmong@gmail.com
<https://www.linkedin.com/in/chapman-mong/>
<https://github.com/chapmanmong>
<https://chapmanmong.me/>

EDUCATION

FinTech Bootcamp, Monash University (2021), Melbourne, Victoria

A 24-week intensive program focused on technical skills in Python, SQL, Big Data, Ethereum and Solidity, Blockchain and Machine Learning

Bachelor of Computer Engineering, University of Melbourne (2012)

Graduated with honours in a course that covered Computer Science as well as Electrical Engineering giving me a well-rounded education on all things computer related

Cheuk Man (Chapman) Mong

SOFTWARE ENGINEER

Australian Citizen
Melbourne

SUMMARY

Software Engineer with 4 years of working experience in Typescript, React, React Native, Node, Python, Django, Postgres, Terraform and AWS cloud technologies. Achievements include spearheading projects such as integration of Kone lifts, My Parcel Locker and Salto Access System with a building management system and building a CRM from scratch as well as a live rolling migration of data and services from old system to new one in my most recent role (Eyerising International). I have also recently built and deployed a full stack application (AWS, Terraform, CI/CD, React, ExpressJs, Github Actions, Vitest, Playwright) which you can check out on jh.chapmanmong.me

EXPERIENCE

SOFTWARE ENGINEER EYERISING INTERNATIONAL 2024 May – 2025 Oct

Software Engineer in a two-man IT/software team supporting a global med-tech startup. Responsibilities include building a new full-stack platform (React/Django/Docker/Terraform/AWS/ Auth0/Airwallex payment gateway) based on existing prototype production system and migrating data from MongoDB to a Postgres relational database. The medical device was an embedded NXP device running Freescale Linux and runs a Qt C++ program which I also maintained/added new features.

Responsibilities and achievements

- Built a full-stack production Frontend/Backend for a portal and backend system which handles business operations as well as communicating with a Linux based medical device running firmware written in C++/QT
- Designed and implemented complex business logic (subscriptions, plans, payment models, distributor/clinic hierarchies, multilanguage, automations, etc.)
- Migration of production MongoDB database to Postgres database and created a synchronization system to keep them in sync during the transition phase
- Created a virtual cross-compilation platform in docker to compile firmware binaries which cut down compilation time by 80% compared to previous workflow
- Working with non-technical colleagues to serve their needs
- Acted as subject matter expert to influence engineering decisions

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- AWS Certified Developer Associate
- AWS Certified Solutions Architect
- Google Project Management Certificate
- IBM Applied DevOps Engineering Professional Certificate

SOFTWARE DEVELOPER CONNEXX/SMART URBAN PROPERTIES AUSTRALIA 2021 Dec – 2024 Apr

Software developer in a building management startup. Responsibilities included adding new features and maintaining a residential building management SaaS system built on Typescript, React, React Native, AWS (DynamoDB, API Gateway, Lambda, etc.). Hired as an IoT and integrations engineer whose main responsibility is integrating with third party applications (Kone lift, My Parcel Locker, access control systems), but responsibilities grew rapidly (subject matter expert on core system/onboarding of new developers/primary contact for customers) due to unexpected change in staff.

Responsibilities and achievements

- Developing new features
- Maintaining and adding proprietary APIs
- Integrating with third party APIs
- Liaising with stakeholders (building managers, external developers, residents)
- Successfully completed new features such as Kone lift and Salto access control integration and timetabling system under tight deadlines
- Moved complicated portal codebase to Vite from Webpack for exponential performance boost

FREELANCE TRANSLATOR REMOTE 2015 - 2021

Chinese -> English translator of general and technical documents including websites, academic papers, press releases, manuals, documentation, and other content. Worked directly and indirectly with many well-known international companies to deliver native-sounding translations and content.

Skills and responsibilities

- Exceptional research and written communication skills
- Ability to meet deadlines and work independently
- Coordination with stakeholders across timezones and cultures

PROJECTS

Job Hunt Tracker <https://jh.chapmanmong.me>

A full stack application with full CI/CD/IaC deployed on AWS

Repository is private but happy to show and tell if requested

Cloud & Infrastructure: AWS (ECS Fargate, RDS, ALB, CloudFront, ECR, Route53, Secrets Manager, IAM, S3), Terraform (IaC, modules, remote state)

Backend: Node.js, Express, TypeScript, REST API design, JWT auth, bcrypt, PostgreSQL, parameterized SQL

Frontend: React 19, TypeScript, Vite, Tailwind CSS v4, shadcnui, React Context API, component-driven architecture

Testing: Vitest unit testing, Playwright end-to-end testing

DevOps: Docker, Docker Compose, GitHub Actions CI/CD, CloudWatch logging

Security: SQL injection prevention, secrets management, token-based auth, network isolation (private subnets, security groups)

Architecture: Full-stack TypeScript, user-isolated data access, RESTful API conventions, modular IaC

PROJECTS (CONTINUED)

Text-to-speech Single Container App

<https://tts.chapmanmong.me>

A single docker container app hosted on own server

Infrastructure: Docker on bare-metal Linux, system-managed container lifecycle, Caddy reverse proxy, Cloudflare DNS & tunnelling

Backend: Python, FastAPI, Edge TTS API, concurrent chunked voice synthesis, SHA256 hash caching

Frontend: Minimal responsive HTML/JS/CSS, task progress polling, mobile-friendly

Architecture: Async task queue with progress tracking, chunked synthesis for long-form text, local MP3 cache with content-addressed storage, concurrency throttling to avoid rate limiting

Observability (planned): Prometheus metrics endpoint (/metrics), structured JSON logging, request latency tracking, cache hit ratio monitoring, Grafana dashboard

Smart thermal jacket (capstone Uni project)

<https://chapmanmong.me/projects/smartthermaljacket.html>

Worked with two electrical engineering final year students to create a self-heating/cooling jacket using Peltier plates. Responsible for programming an Arduino microcontroller (in C++) to create a user interface (buttons and LCD screen) and the logic for controlling 6 Peltier plates and the voltage for heating and cooling